In terms of computer networks then separate standards have been developed by each of these two methods. The standards organization manner of developing standards has resulted in the ISO model while the (approximately) de facto manner has resulted in the development of TCP/IP.

2.2.2 ISO MODEL

The International Standards Organization (ISO), based in Geneva Switzerland, is composed of groups for various countries that set standards working towards the establishment of world-wide standards for communication and data exchange. One notable accomplishment has been the development of a reference model that contains specifications for a network architecture for connecting dissimilar computers, with a main goal being that of producing an open and non-proprietary method of data communication. This reference model, called the Open Systems Interconnect Reference Model (OSI RM), was developed in 1981 (?) and revised in 1984. The OSI RM uses 7 layers, each independent of each other, to allow computers to exchange data. To transfer a message from user A to user B, the data has to pass through the 7 layers on user's A machine, before being transmitted through the selected medium. At the receiving computer of user B, the data must then pass through the 7 layers again, this time in reverse sequence, before being received by user B. For data to be transferred, it must pass through all 7 layers on both computers. Each layer follows a relatively strict specification and this allows the differing layers to be produced and implemented by different concerns. Each layer can then interface with its neighboring layers even though they may have been developed by different groups. One way of viewing the activities of the layers is that as the original message passes down the layers towards the medium on the computer of user A, additional information, for both formatting or addressing information, is added to the beginning or end of the message. The additional information is added to ensure correct communication. At the other end (i.e. at user B) this information is gradually stripped off at the data passes through the 7 layers - this time in reverse order. The layers are arranged in order as follows: Layer 7, Application Layer. This layer defines network applications such as error recovery, flow control and network access. Note that user applications are not part of the layers. Layer 6, Presentation Layer. This layer determines the format used to exchange data in such aspects as data translation, encryption and protocol conversion. The data from user A is translated to a common syntax that can be understood by user B. In this way it specifies how applications can connect to the network and to user B. Layer 5, Session Layer. This layer controls the communication session between computers. It is responsible for establishing and removing communication sessions between computers. Additional address translations and security are also performed. This layer therefore instigates a data transfer session between user A and user B so that an extended data transfer can take place. Layer 4, Transport Layer. This layer is responsible for ensuring that data is delivered free of error and provides some flow control. This layer ensures that data is transferred as part of the session instigated by the Session Layer. Layer 3, Network Layer. This layer handles the delivery of data by determining the route for the information to follow. The data is divided into packets with addressing information attached. It also translates address from names into numbers. Intermediate addresses are also attached. Layer 2, Data Link Layer. This layer defines the network control mechanism and prepares the packets for transmission. Layer 1, Physical Layer. This layer is concerned with the transmission of binary data between stations and defines the connections. The connection definition includes such aspects as mechanical, electrical,, topology and bandwidth aspects. At the receiving end the process is reversed so that the binary data that is received is translated back into the original message for User B. Note that each layer communicates only with it's immediate neighbors. For example the Transport Layer only communicates with the

Session Layer and with the Network Layer. Each network architecture can be defined by a set of protocols for each of the layers. This allows for a degree of simplification and modularity in design. In spite of an enormous amount of worked and effort having been expended on the ISO RM, very little of it is in use compared to TCP/IP (described in the following section).